

BKN AI Capital — Return Due Diligence

Quantitative hedge-fund CIO perspective • Probability of consistent 5–10% monthly returns over 12 months, after Zerodha live-data integration and before further major feature development.

Evidence-based. No optimism. Trading performance is neither guessed nor invented; base-rate speculation is labeled as such.

P(5–10%/mo, 12 mo)	~2%	OWN-CAPITAL ALLOCATION	0%
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Framing (determines the answer)

5–10% *monthly*, sustained 12 months, is roughly **80–214% annualized** — an all-time-outlier bar, not a "good strategy" bar. Renaissance Medallion, the best track record on record, ran ~39% net / ~66% gross annualized. A retail-scale, hand-coded technical-rule system clearing this *consistently* would be among the best quant results ever. The base rate for that outcome, for this class of system, rounds to ~zero before reading any code.

Threshold statement (evidence sufficiency)

The current evidence is insufficient to estimate this system's returns with confidence. Verified in-repo: **zero live trades, zero out-of-sample backtests, empty strategy statistics** (a strategy needs MIN_SAMPLES=30 to be "proven"; current samples = 0), and **no demonstrated slippage/commission/impact modeling** in the backtester. The only "performance" figures in the codebase are hand-built unit-test fixtures verifying arithmetic (e.g. win_rate==1.0 from one winning paper trade; expectancy_r==1.0 from a 3-trade toy) — not measurements of trading. I therefore give a **base-rate estimate for this class of system, not a measurement of this software.**

1. Estimated probability — ~2% (range 1–4%)

Dominated by base rates: (a) the return bar is outlier-tier; (b) technical rule systems show near-zero-to-negative expectancy after costs in liquid markets; (c) SEBI's 2023 study found ~89–93% of individual Indian derivatives traders lose money. ~2% is essentially the chance that hand-tuned heuristics accidentally encode a large, persistent, cost-surviving edge.

2. Expected monthly return range (base-rate speculation, NOT measured)

Scenario	Monthly net	Rationale
Worst realistic	-10% to -20%	Costs + adverse selection; far worse on leveraged F&O (leverage compounds losses).
Base case	-2% to +1%	Breakeven-to-negative after fees/slippage — empirical default for uncalibrated technical systems.
Optimistic	+1% to +3%	Only if a real edge exists and survives costs — still well below the 5–10% target.

I explicitly refuse to output a "5–10%" base case: no evidence supports it.

3. Maximum expected drawdown

• Cash equity, unlevered: ~15–30%.

• Leveraged / F&O: ~35–60%+, with non-trivial risk-of-ruin.

• Actual DD is undefined without live data; these are class base rates. The risk gates (NO-TRADE, sizing, invalidation) *should* compress the left tail but are unvalidated live.

4. Confidence level

• **Low** confidence in any specific return point estimate (no data to calibrate).

• **High** confidence in the direction: consistent 5–10%/month is extremely unlikely. The two are kept separate.

5. Strongest evidence supporting success

- **Real risk discipline in code**, not slideware: hard risk gates, a genuine NO-TRADE verdict (composite floors, hostile-regime stand-aside), position sizing, invalidation levels. Poor risk control is the #1 killer of retail returns; this structurally avoids the worst behaviors.
- **Explainable, auditable decisions** (cited findings per agent) — makes edge testable and failures diagnosable.
- **Clean engineering** enables rigorous validation quickly — leverage on the research problem, though silent on current edge.

6. Strongest evidence suggesting failure

- **No validated edge anywhere**. Signals are deterministic hand-tuned rules (no ML/learning); strategy stats empty; no OOS/walk-forward result committed.
- **No demonstrated cost realism** in the backtester — a technical system's thin gross edge typically dies on costs.
- **High, uncontrolled overfitting risk**: thresholds ($RR \geq 2$, RSI bands, $ADX \geq 25$, composite floors) hand-chosen with no multiple-testing / deflated-Sharpe discipline.
- **Outlier-tier target** against brutal base rates for the exact population (Indian retail technical trading).

7. Assumptions doing the most work

- **Edge existence (dominant)**: I assume ~zero persistent net edge because none is demonstrated. If wrong, the whole estimate moves — and it is the only thing that matters.
- **Cost erosion**: realistic slippage/fees materially erode gross performance.
- **Instrument choice**: cash equity (lower ceiling, lower ruin) vs leveraged F&O (higher variance, higher ruin) — unspecified, swings the tails.
- **The 5–10% bar** is outlier-tier (well supported, low uncertainty).

8. Evidence that would most increase confidence (in order)

- **≥ 200 –300 live forward paper trades on real Zerodha data with statistically significant positive expectancy AFTER realistic costs** (t-stat on per-trade R, not just a positive sum). Highest-value single artifact.
- **OOS + walk-forward backtests over ≥ 3 –5 years and multiple regimes** (2018 grind, 2020 crash/recovery, 2022 drawdown) with deflated Sharpe / multiple-testing correction.
- **Live-vs-backtest reconciliation** proving the simulator isn't optimistic.
- **Per-strategy capacity & cost sensitivity** (does edge survive at size and real slippage?).
- **Parameter-stability maps** showing thresholds aren't knife-edge overfits.

Without at least the first two, any return claim is speculation.

9. Own-capital allocation on current software: 0%

A quant CIO does not allocate risk capital to a strategy with no validated expectancy, no live track record, and an outlier target — regardless of code quality. Code quality is table stakes, not edge. I would fund an engineering/paper *research budget*, but that is R&D spend, not an investment allocation. **Off 0% → 1%** requires a clean cost-aware OOS backtest **plus** ~200 live paper trades with significant positive expectancy. **5%+** requires months of live, real-money-shadowed performance with controlled drawdown.

10. Six-month priorities to maximize the chance (ordered)

1. **Cost-aware validation harness** (realistic slippage/fees/impact; F&O specifics) + OOS & walk-forward per strategy across regimes. **Kill any strategy without a statistically significant, cost-surviving edge**. Nothing else matters until this exists.

2. **Live forward paper trading on real Zerodha data at scale** (hundreds of trades); measure realized expectancy; reconcile live vs simulated fills.
3. **Overfitting controls:** deflated Sharpe, multiple-testing correction on hand-tuned thresholds, parameter-stability analysis.
4. **Execution & capacity modeling:** edge degradation with size/spread/adverse selection; set instrument scope (equity vs F&O) deliberately.
5. **Risk-of-ruin / drawdown validation** and sizing calibration under live conditions; verify the NO-TRADE and invalidation gates actually cap the left tail.

Bottom line: The evidence base is currently insufficient to credibly forecast returns, and against the correct base rates a consistent 5–10%/month over 12 months is highly improbable (~2%). The software today is a well-built, disciplined research and paper-trading platform with **no demonstrated alpha** — a vehicle for *discovering* whether an edge exists, not evidence that one does. Allocation on the current software: **0%**. Only validated, cost-aware, out-of-sample and live expectancy data changes that.